Suggested Medium titles (pick one)

1) **Stop Copying Rows: The “Keep/Ignore Note” Behind Fast Databases (SV vs Bitmaps)** 2) **How Databases Go Fast (Without Magic): Selection Vectors vs Bitmaps** 3) **Vectorized Execution Explained for Beginners: The Filter Trick That Saves CPU**

Stop Copying Rows: The “Keep/Ignore Note” Behind Fast Databases (SV vs Bitmaps)

Picture this:

You run a query on a huge table. It finishes in seconds.

Your brain says: “Wow, that database must be doing something genius.”

The truth is often simpler:

> Fast engines avoid doing the same heavy work again and again. Especially: they try to avoid copying data around.

This article explains one surprisingly powerful idea from the research paper **“Filter Representation in Vectorized Query Execution” (Ngom et al., 2021)**:

> Instead of physically removing rows after every filter, engines keep data in place and carry a small **keep/ignore note**.

That keep/ignore note is called a **filter representation**.

The one-sentence meaning

A **filter representation** is just a way to remember:

> “In this small group of rows, which ones should I keep using, and which ones should I ignore?”

Quick definitions (easy)

- **Row**: one record (example: one order).

- **Batch**: a small group of rows processed together (example: 1,000 rows).
- **Pipeline**: steps in a row (scan → filter → compute → sum).
- **Filter**: a rule that keeps some rows and rejects others.
- **Filter representation**: the keep/ignore note attached to a batch.

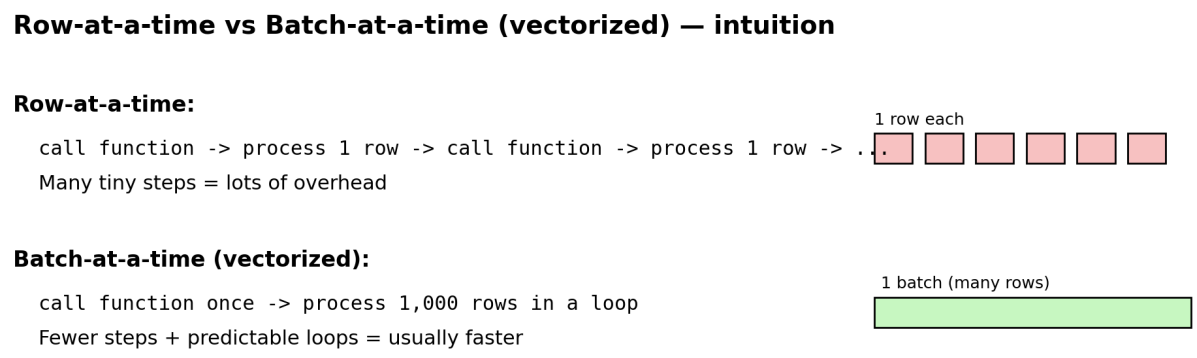
Why databases process in batches (vectorized execution)

Computers pay a cost every time they “start doing something.” If you do work one row at a time, you pay that start-cost a lot.

So many analytics engines do this instead:

- grab a **batch** (like 1,000 rows)
- loop over it in a tight, predictable way

Diagram: row-at-a-time vs batch-at-a-time

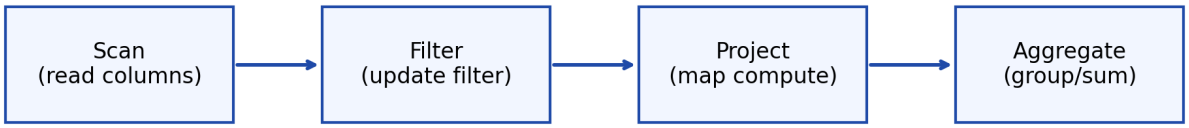


The big picture (architecture)

A vectorized database processes batches through operators (think: stations on a factory line).

Diagram: batch + filter flows through the pipeline

Vectorized execution: operators pass a batch + a filter through the pipeline



Each operator processes a BATCH (vector) + carries a FILTER (SV or BM)
so it knows which positions are still valid without copying rows.

Important idea:

- The batch values (columns) can stay in memory.
- The keep/ignore note changes as you apply filters.

Why this filter thing exists (the purpose)

When you filter data, you have two choices.

Option A: Copy only survivors

You build a new output batch containing only rows that passed.

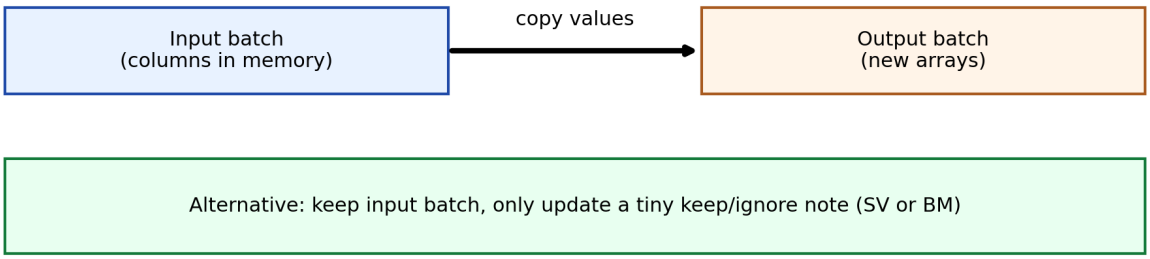
This is simple, but copying again and again can be expensive.

Option B: Don’t copy; keep the batch and carry a keep/ignore note

You keep the values where they are. You only update the small note that says which positions are valid.

Diagram: why copying can be expensive

Why copying survivors can be expensive (memory traffic)



Copying moves lots of bytes. Updating SV/BM moves very little.

This is the mindset shift:

- Copying moves lots of data.
- Updating a small note moves very little.

A tiny example with real numbers (8 rows)

We have one batch with 8 rows.

Positions: `0 1 2 3 4 5 6 7`

`amount`: `20 150 5 130 200 10 90 180`

Filter rule:

> Keep rows where amount > 100

Survivors are positions: **1, 3, 4, 7**

Diagram: what changes after the filter

Example query step: WHERE amount > 100 (batch of 8)

Batch positions: `0  1  2  3  4  5  6  7`

amount column: `20 150  5 130 200 10 90 180`

predicate >100: `F  T  F  T  T  F  F  T`

Selection vector: **[1, 3, 4, 7]**

Bitmap: **0 1 0 1 1 0 0 1**

Next operator (e.g., compute tax = amount*1.2) can either:

- Selective: compute only for positions in SV
- Full: compute for all 0..7, then keep results where BM=1

Two ways to write the keep/ignore note

Both options mean the same thing (“keep 1,3,4,7”), they’re just stored differently.

1) Selection Vector (SV) = list of survivor positions

- `SV = [1, 3, 4, 7]`

How to read it:

- “Only look at rows number 1, then 3, then 4, then 7.”

Real-life analogy:

- a guest list of seat numbers for people allowed inside.

2) Bitmap (BM) = 0/1 flags for each position

- `BM = 0 1 0 1 1 0 0 1`

How to read it:

- “At each position: 1 means keep it, 0 means ignore it.”

Real-life analogy:

- a row of light switches (ON = keep, OFF = ignore).

Real example: one query, step-by-step (what a DB is doing)

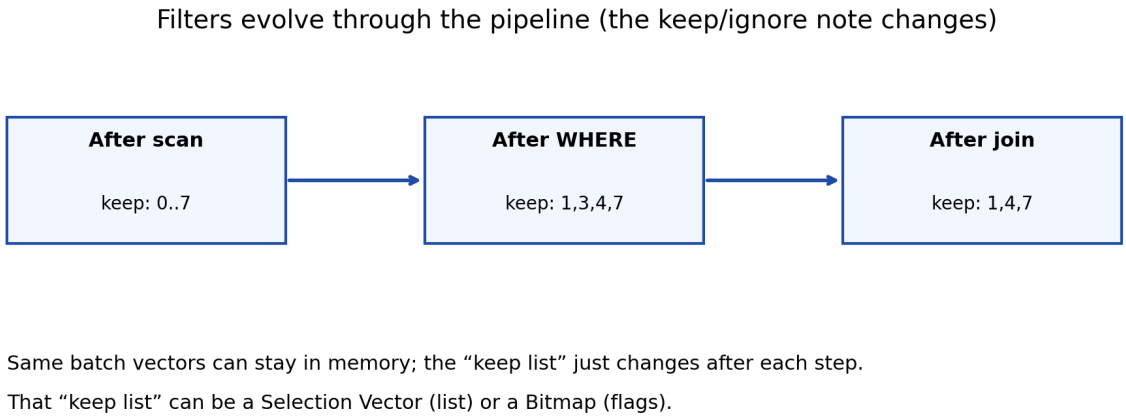
Imagine a normal analytics query:

```sql
SELECT country, SUM(amount * 1.2) AS taxed_revenue
FROM orders
WHERE amount > 100
GROUP BY country;
```

Conceptually:

1) **Scan**: read columns in batches (`amount[]`, `country[]`). 2) **Filter**: apply `amount > 100` → update keep/ignore note (SV or BM). 3) **Compute**: calculate `amount * 1.2` for valid rows. 4) **Aggregate**: group by `country`, sum values for valid rows.

Diagram: the keep/ignore note changes across steps



Same batch vectors can stay in memory; the “keep list” just changes after each step.

That “keep list” can be a Selection Vector (list) or a Bitmap (flags).

Why SV vs BM matters (the easy performance story)

After filtering, the next step might compute something. Example: `amount * 1.2`.

There are two styles:

Style A: Compute only survivors (often pairs well with SV)

If only a few rows survived, don’t waste time computing the rest.

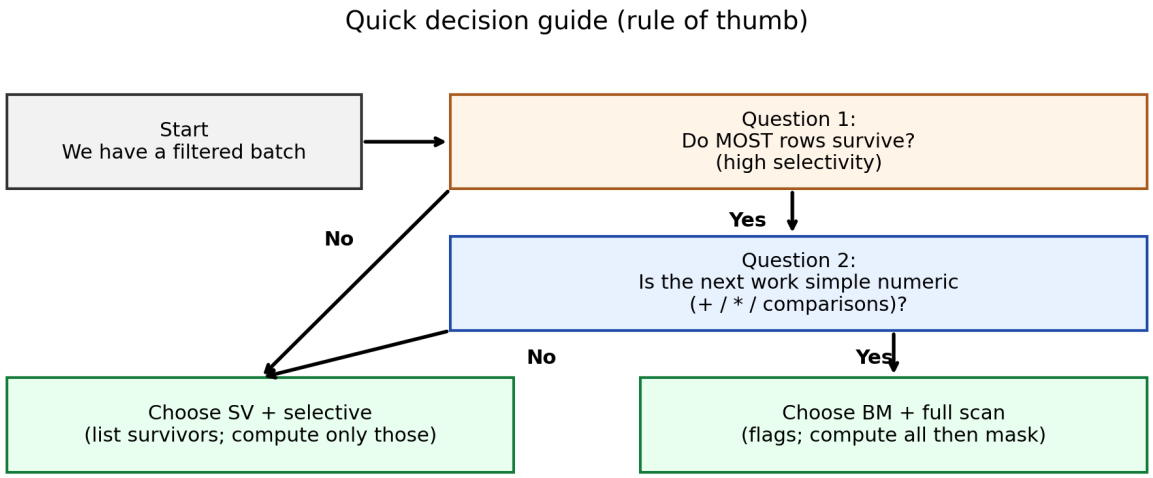
Style B: Compute everything, then ignore bad rows (often pairs well with BM)

This can be faster when:

- most rows survived, and
- the work is simple math,
- the computer can run a very tight loop efficiently.

Cheat-sheet: what to pick (rule of thumb)

Diagram: quick decision flow



Or a table:

| If… | Usually choose… | Why | |---|---|---| | Only a few rows survive | SV | Touch only survivors | | Most rows survive and work is simple numeric (math/compare) | BM | Scanning everything can be very efficient | | Work is complicated/irregular (strings, complex logic) | SV | Less overhead than many flags |

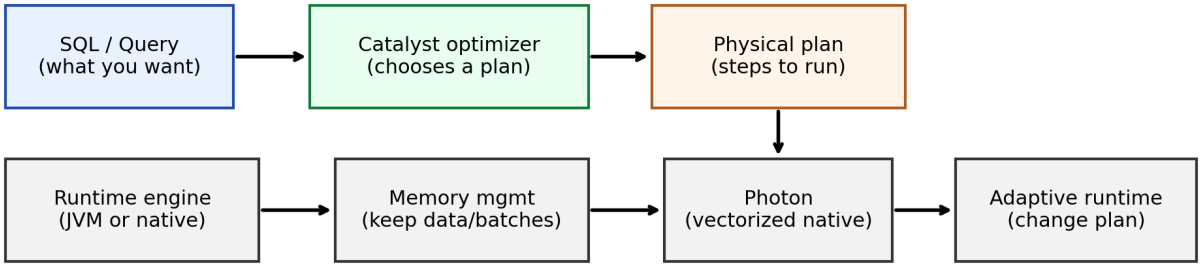
Real-world connection: Spark / Databricks (simple)

Spark and Databricks are big systems, but they chase the same goal:

> Do less wasted work, move less data, and use CPU/memory efficiently.

Diagram: simple view of the stack

How this relates: optimizer chooses steps, runtime executes in batches + manages memory



Memory management

Memory management is how the system uses RAM so it doesn't crash and doesn't slow down.

Link to our story:

- lots of copying → lots of memory work
- keeping data in place → less memory work

Catalyst optimizer (Spark)

Catalyst is Spark's "planner."

It tries to pick a plan that makes the runtime cheaper, like:

- pushing filters early
- simplifying expressions

Photon (Databricks)

Photon is an execution engine that is very good at running batch-style operations efficiently.

This connects directly to the paper's world: vectorized execution loves predictable loops.

Runtime adaptivity / dynamic query optimization

Meaning:

> The system can change its plan while running, based on what it learns.

Example:

- the planner expects 80% of rows survive

- at runtime only 2% survive
- the engine switches strategies to match reality

Partition coalescing

A partition is a chunk of data processed in parallel.

Coalescing means:

- too many tiny chunks → merge into fewer, bigger chunks

It's the same principle as batching:

- fewer tiny tasks → less overhead

Common beginner confusions (quick fixes)

"Why not just delete the bad rows?"

Because deleting/copying all the time costs work. It's often cheaper to leave data where it is and just remember "ignore these positions."

"Is SV always better?"

No. If most rows survive and the next step is simple math, BM + full scan can be faster.

"Is BM always better?"

No. For irregular work (strings, complex rules), SV can be simpler and faster.

A tiny checklist you can use in real projects

If you're tuning a Spark/Databricks job or any analytics pipeline:

- **Are you copying/shuffling data a lot?** (often expensive)
- **Can you filter earlier?** (reduce wasted work)
- **Is your work simple numeric or complex/branchy?** (impacts what optimizations help)
- **Are you doing too many tiny tasks?** (coalesce/adjust partitioning)

Conclusion

If you only remember one thing:

- Fast engines often don’t copy data after every filter.
- They keep batches in place and carry a small keep/ignore note.
- That note can be a **Selection Vector (list of survivors)** or a **Bitmap (0/1 flags)**.

And the broader lesson:

> Performance is often about reducing overhead and reducing unnecessary movement of data.